РОСТОВСКИЙ ЦСМ

Federal Agency
on Technical Regulating and Metrology
Federal budget institution
«STATE REGIONAL CENTRE OF STANDARDIZATION, METROLOGY AND TESTING IN ROSTOV REGION»
LABORATORY FOR FOOD, AGRICULTURAL PRODUCTS AND CONSUMER GOODS
(FBI ROSTOV CSM)

№ RA.RU.21ПЛ84

AFFIRM
Head of Laboratory
_____ A.A. Starikov
« 19 » August 2025

ANALYSIS REPORT №15515-25

Product sample name*:	Russian origin barley, in bulk. Crop year 2025. seal 0025690
Name and address of the company*:	"Agrokontrol" LLC, 347900, Russia, Rostov Region, Taganrog, Admirala Kryuysa str., 27
Delivery date:	05.08.2025
Period of analysis:	05.08 – 11.08.2025
Transport vehicle name*:	m/v "DRY VOYAGER 1"
Shipment quantity*:	5000 +/- 10% MT
Port terminal*:	---
Subcontractor:	---
Loading port*:	Taganrog

ANALYSIS RESULTS

NORMATIVE DOCUMENTS AND MEASUREMENT INSTRUMENTATION

Methodological Instructive Regulations "Identification and isomer-specific determination of poly-CDD and poly-CDF in meat, bird, fish and their by-products, and other fat-containing products by chromato-mass-spectrometry", 1999
Methodological Instructive Regulations 4.1.1023-01. "Isomer-specific determination of polychlorinated biphenyls (PCBs) in food", 2001.
EC 152/2009

Standard Samples: СОП 0501-03 – 0520-03
СОП 0601-03 – 0606-03
BCR – 614 Solution S8
CM-220, WHO PCB Mixture

Poisonous material	Toxic equivalent factor (TEF)	Found concentration	
		ng/kg	TEQ, ng/kg
1	2	3	4
Dibenzo-p-dioxins (PCDDs)			
2,3,7,8-TetraCDD	1	< 0,025	< 0,025
1,2,3,7,8-PentaCDD	1	< 0,025	< 0,025
1,2,3,4,7,8-HexaCDD	0,1	< 0,050	< 0,005
1,2,3,6,7,8- HexaCDD	0,1	< 0,050	< 0,005
1,2,3,7,8,9- HexaCDD	0,1	< 0,050	< 0,005
1,2,3,4,6,7,8- HeptaCDD	0,01	< 0,100	< 0,001
OctaCDD	0,0003	< 0,250	< 0,000075
Dibenzofurans (PCDFs)			
2,3,7,8- TetraCDF	0,1	< 0,025	< 0,0025

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1	2	3	4
1,2,3,7,8- PentaCDF	0,03	< 0,025	< 0,00075
2,3,4,7,8- PentaCDF	0,3	< 0,025	< 0,0075
1,2,3,4,7,8- HexaCDF	0,1	< 0,050	< 0,005
1,2,3,6,7,8- HexaCDF	0,1	< 0,050	< 0,005
1,2,3,7,8,9- HexaCDF	0,1	< 0,050	< 0,005
2,3,4,6,7,8- HexaCDF	0,1	< 0,050	< 0,005
1,2,3,4,6,7,8- HeptaCDF	0,01	< 0,100	< 0,001
1,2,3,4,7,8,9- HeptaCDF	0,01	< 0,100	< 0,001
OctaCDF	0,0003	< 0,250	< 0,000075
OtherTetrachlordibenzodioxins	-	< 0,025	-
OtherPentachlordibenzodioxins	-	< 0,025	-
OtherHexachlordibenzodioxins	-	< 0,050	-
OtherHeptachlordibenzodioxins	-	< 0,100	-
OtherTetrachlordibenzofurans	-	< 0,025	-
OtherPentachlordibenzofurans	-	< 0,025	-
OtherHexachlordibenzofurans	-	< 0,050	-
OtherHeptachlordibenzofurans	-	< 0,100	-
Dioxin-like PCBs Non-ortho PCBs			
PCB – 77	0,0001	< 1,000	< 0,0001
PCB – 81	0,0003	< 1,000	< 0,0003
PCB - 126	0,1	< 1,000	< 0,1
PCB - 169	0,03	< 2,000	< 0,06
Dioxin-like PCBs Mono-ortho PCBs			
PCB - 105	0,00003	< 1,000	< 0,00003
PCB - 114	0,00003	< 1,000	< 0,00003
PCB – 118	0,00003	< 1,000	< 0,00003
PCB - 123	0,00003	< 1,000	< 0,00003
PCB - 156	0,00003	< 2,000	< 0,00006
PCB - 157	0,00003	< 2,000	< 0,00006
PCB - 167	0,00003	< 2,000	< 0,00006
PCB - 189	0,00003	< 4,000	< 0,00012
Threshold of detectability 2,3,7,8- TetraCDD -^{13}C, ng/kg: 0,05	**Total concentrationin TEQ**		**< 0,259**
non Dioxin-like PCBs			
PCB – 28	---	< 1,000	---
PCB – 52	---	< 1,000	---
PCB – 101	---	< 1,000	---
PCB - 138	---	< 1,000	---
PCB – 153	---	< 1,000	---
PCB - 180	---	< 1,000	---

ng WHO-PCDD/F- TEQ/kg (88% d.w.)=0.098. Maximum level 0.75 ng WHO-TEQ/kg

ng WHO-PCDD/F/PCBs- TEQ/kg (88% d.w.)=0.259. Maximum level 1.25 ng WHO-TEQ/kg

Dioxin and Dioxin-like PCB's <1.25 ng WHO-PCDD/F-PCB-TEQ/kg in compliance with EU Regulation 277/2012

The sum of non- dioxin like PCB's of the above mentioned sample is below the maximum level of 10 µg/kg relative to a feed with a moisture content of 12 %

Remarks: «-» in «Concentration» column means that the actual value does not exceed the detection limit of the method

Tables of TEFs for risk assesment based on EU Regulations 277/2012

WHO (PCDD/PCDF) incl. 0.5*LOD, ng/kg TEQ	<	0.05
WHO (PCDD/PCDF) incl. 1.0*LOD, ng/kg TEQ	<	0.10
WHO (Dl PCB) incl. 0.5*LOD, ng/kg TEQ	<	0.08
WHO (Dl PCB) incl. 1.0*LOD, ng/kg TEQ	<	0.16
WHO (PCDD/PCDF + Dl PCB) incl. 0.5*LOD, ng/kg TEQ	<	0.13
WHO (PCDD/PCDF +Dl PCB) incl. 1.0*LOD, ng/kg TEQ	<	0.26

LABORATORY RESULTS

based on the analytical results, we confirm that Total concentration in TEQ of PCDD/F + Dioxin-like PCB not exceed the levels indicated in EC/2002/32, EC/2023/915, EU Regulation 277/2012 and its Ammendments

Responsible for the preparation and execution of the test report:

Engineer		A.A. Ishchenko
(Job position)	(Signature)	(Surname)

End of analysis report